PROBLEMS IN DIFFERENTIATION

Limits, L'Hospital's Rule & Differentiability

PART 1

(a) $\lim_{x \to \infty} \dfrac{x - \sin x}{2x + \sin x}$

(b) $\lim_{x \to \infty} \dfrac{2x + \sin 2x + 1}{(2x + \sin 2x)(\sin x + 3)^2}$

(c) $\lim_{x \to 0^+} \left(2 \sin \sqrt{x} + \sqrt{x} \sin \dfrac{1}{x}\right)^x$

(d) $\lim_{x \to 0} \left(1 + x e^{-\frac{1}{x^2}} \sin \dfrac{1}{x^4}\right)^{e^{-\frac{1}{x^2}}}$

[SOLUTION (Ravi Shankar)]

PART 2

Is the function given by

$$f(x) = \begin{cases} \dfrac{1}{x \ln 2} - \dfrac{1}{2^x - 1}, & x \neq 0, \\ \dfrac{1}{2}, & x = 0 \end{cases}$$

differentiable at zero?

[SOLUTION (Ravi Shankar)]

PART 3

Prove that if f is twice continuously differentiable on $\mathbb{R}$ such that $f(0) = 1$, $f'(0) = 0$, and $f''(0) = -1$, then for $a \in \mathbb{R}$,

$$\lim_{x \to +\infty} \left(f\left(\frac{a}{\sqrt{x}} \right) \right)^x = e^{-\frac{a^2}{2}}.$$

Hint: Take logarithm and use Taylor expansion of $f(t)$ about $t = 0$ up to the second-order term.

PART 4

For $a > 0$, $a \neq 1$, evaluate

$$\lim_{x \to +\infty} \left(\frac{a^x - 1}{x(a-1)} \right)^{\frac{1}{x}}.$$

Hint: Take logarithm of the expression. Consider separately the cases $0 < a < 1$ and $a > 1$, and identify the dominant term in $a^x - 1$.

PART 5

Verify the following claims:

(a)

$$\lim_{x \to 1^-} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = \ln 2.$$

Hint: Use the power series $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$.

(b)

$$\lim_{x \to 1} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^x} = \ln 2.$$

Hint: Put $x = 1$ and use the alternating harmonic series.

(c)

$$\lim_{x \to 1^-} \sum_{n=1}^{\infty} (x^n - x^{n+1}) = 1.$$

Hint: First take the partial sum. Notice that the series is telescoping.

(d)

$$\lim_{x \to 0^+} \sum_{n=1}^{\infty} \frac{1}{2^n n^x} = 1.$$

Hint: As $x \to 0^+$, $n^x \to 1$. Then use the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$.

(e)

$$\lim_{x \to \infty} \sum_{n=1}^{\infty} \frac{x^2}{1 + n^2 x^2} = \frac{\pi^2}{6}.$$

Hint: Divide numerator and denominator by x^2 and use $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

PART 6

Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges. Find

$$\lim_{x \to 1^-} \sum_{n=1}^{\infty} a_n x^n.$$

Hint: Apply **Abel's theorem** for power series.

Think • Apply • Solve • Master